

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 15: Separation & Wake Turbulence

Syllabus Source: CAE Oxford (pp. 297-330) | Pass Target: >= 90% AviationExam

1. VERTICAL SEPARATION & RVSM

- Standard Vertical Separation: 1,000 ft below FL 290; 2,000 ft above FL 290.
- RVSM (Reduced Vertical Separation Minimum): 1,000 ft vertical separation between FL 290 and FL 410 inclusive for approved aircraft. Above FL 410, separation is 2,000 ft.

2. HORIZONTAL RADAR & PROCEDURAL SEPARATION

- Standard Radar Separation: 5 NM en-route; 3 NM in terminal areas (reduced to 2.5 NM under special radar criteria).
- Non-Radar Longitudinal Separation: 15 minutes standard; reduced to 10 minutes if navigation aids permit frequent position determination; 5 minutes for climbing/descending aircraft on same track; 3 minutes for departing aircraft.
- DME Separation: 20 NM on same track; 10 NM if leading aircraft maintains >= 20 kt higher true airspeed.
- Mach Number Technique: Used in oceanic/continental airspace: 80 NM or 10 minutes when longitudinal separation is based on Mach number.

3. NON-RADAR LATERAL SEPARATION BY TRACK DIVERGENCE

Lateral separation between two aircraft departing or navigating along diverging tracks: - VOR: Tracks diverge by at least 15° and at least one aircraft is at or beyond 15 NM from the VOR. - NDB: Tracks diverge by at least 30° and at least one aircraft is at or beyond 15 NM from the NDB. - Dead Reckoning (DR): Tracks diverge by at least 45° and at least one aircraft is at or beyond 15 NM from intersection.

4. WAKE TURBULENCE CATEGORIES & MINIMA MATRIX

Category	Certified Take-Off Mass (MTOM)	Radar Wake Separation (Trailing)
SUPER	Airbus A380-800	Behind Super: Heavy 6 NM, Medium 7 NM, Light 8 NM.
HEAVY (H)	136,000 kg or more	Behind Heavy: Heavy 4 NM, Medium 5 NM, Light 6 NM.
MEDIUM (M)	Between 7,000 kg and 136,000 kg	Behind Medium: Heavy/Medium Radar min, Light 5 NM.
LIGHT (L)	7,000 kg or less	Light twins and single-engine aircraft.

[!] EXAM TRAP (AVIATIONEXAM) - Timed Departure Wake Turbulence Separation (Non-Radar)

When taking off behind a HEAVY aircraft from the same runway: - Full length of runway: 2 MINUTES separation. - From an INTERMEDIATE intersection (displaced threshold): 3 MINUTES separation (due to vortex drift).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the maximum certified take-off mass for a HEAVY wake turbulence category aircraft?

- [A]: 100,000 kg or more
- [B]: 136,000 kg or more
- [C]: 150,000 kg or more
- [D]: 300,000 kg or more

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The HEAVY wake turbulence category applies to aircraft of 136,000 kg (300,000 lbs) or more.

Q2: What is the radar wake turbulence separation for a LIGHT aircraft following a HEAVY aircraft on approach?

- [A]: 4 NM
- [B]: 5 NM
- [C]: 6 NM
- [D]: 7 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Heavy behind Heavy is 4 NM; Medium behind Heavy is 5 NM; Light behind Heavy is 6 NM.

Q3: When departing behind a Heavy aircraft from an intermediate intersection, what is the required time separation?

- [A]: 2 minutes
- [B]: 3 minutes
- [C]: 4 minutes
- [D]: 5 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. From an intermediate intersection, 3 minutes separation is required (compared to 2 minutes from full length).

Q4: What is the standard non-radar longitudinal separation between two aircraft on the same track if navigation aids are not available?

- [A]: 5 minutes
- [B]: 10 minutes
- [C]: 15 minutes
- [D]: 20 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Standard procedural separation is 15 minutes, reduced to 10 minutes when nav aids permit frequent updates.

Q5: What is the vertical separation applied in RVSM airspace between FL 290 and FL 410?

- [A]: 500 ft
- [B]: 1,000 ft
- [C]: 2,000 ft
- [D]: 4,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Reduced Vertical Separation Minimum (RVSM) provides 1,000 ft vertical separation between FL 290 and FL 410.

Q6: What is the radar wake turbulence separation for a MEDIUM aircraft following a HEAVY aircraft on approach?

- [A]: 3 NM
- [B]: 4 NM
- [C]: 5 NM
- [D]: 6 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Heavy behind Heavy is 4 NM; Medium behind Heavy is 5 NM; Light behind Heavy is 6 NM.

Q7: What is the minimum lateral divergence angle required between two departing aircraft navigating via VOR?

- [A]: 10°
- [B]: 15° (at or beyond 15 NM)
- [C]: 30°
- [D]: 45°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Lateral separation by VOR requires tracks diverging by at least 15° at a distance of at least 15 NM from the facility.